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**National Institute of Technology, Raipur (Chhattisgarh), India**

Department of Information Technology

**Minor Project Final Presentation — November 2026**

# Spatial-Guided Spectral-Channel Fusion Network for Hyperspectral Remote Sensing Image Analysis

**Under the Guidance of**

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B.Tech Seventh Semester, Nov 2026

National Institute of Technology, Raipur

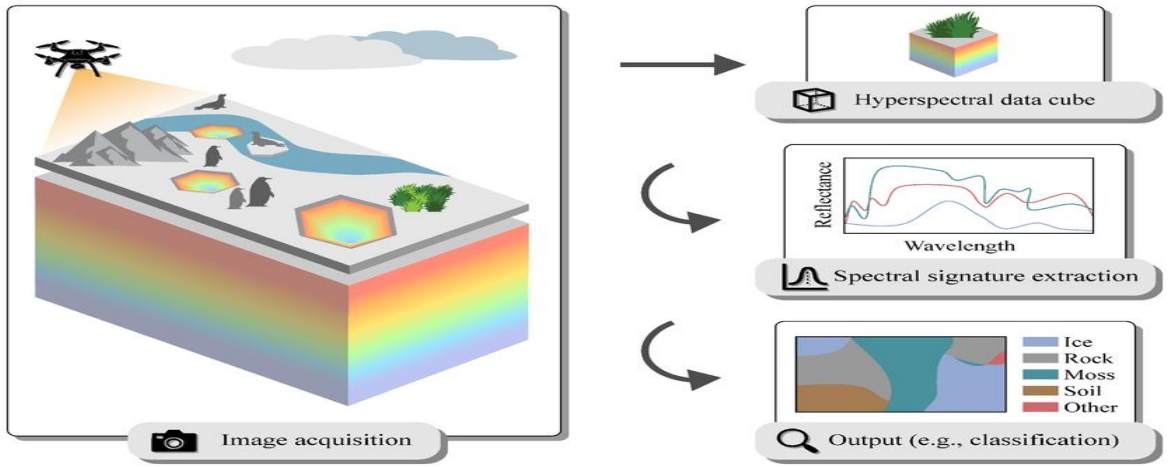
# Outline

- Recap: Problem Statement & Research Objectives
- SGSCF-Net Architecture Overview
- Module Design: SBAM, CBSA, SCFG
- Hybrid Loss Function
- Experimental Setup & Benchmark Datasets
- Results: Indian Pines, Pavia University, Salinas Valley
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- Data Scarcity Stress Testing (1%–10%)
- Cross-Dataset Summary & Key Achievements
- Future Work & Research Directions

# Recap: Problem Statement & Research Objectives

## Problems Identified (Mid-Semester)

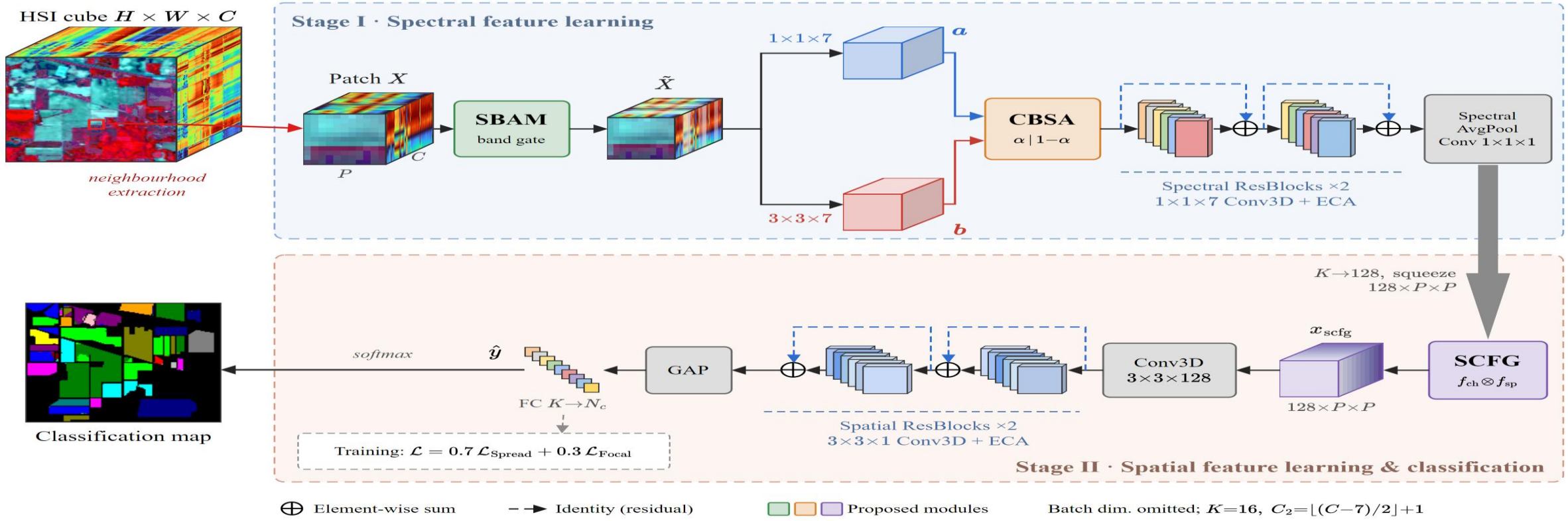
- Hughes Phenomenon: High bands + low samples = overfitting
- Memory Explosion: 3D-CNNs & Transformers exceed 85M+ parameters
- Feature Entanglement: Spatial + spectral mixing blurs class signatures
- Class Imbalance: Minority classes (Oats: 20 samples) collapse to 0%
- Unfiltered Noise: Atmospheric absorption corrupts raw band features




## What We Delivered (This Presentation)

- Fully implemented & trained SGSCF-Net across 3 benchmark HSI datasets (Indian Pines, Pavia University, Salinas Valley)
- 120+ GPU training runs: 60-run patch/batch grid search + 12 data scarcity stress tests + 24 baseline benchmark runs
- Competitive or superior accuracy vs. 8 SOTA baselines while using only 308K parameters (vs. 6.4M for SSAN, ~610K for ViT)
- Complete 31-page ACM-format research paper with per-class breakdowns, classification maps, ablation studies, convergence analysis

# SGSCF-Net: End-to-End Architecture Overview



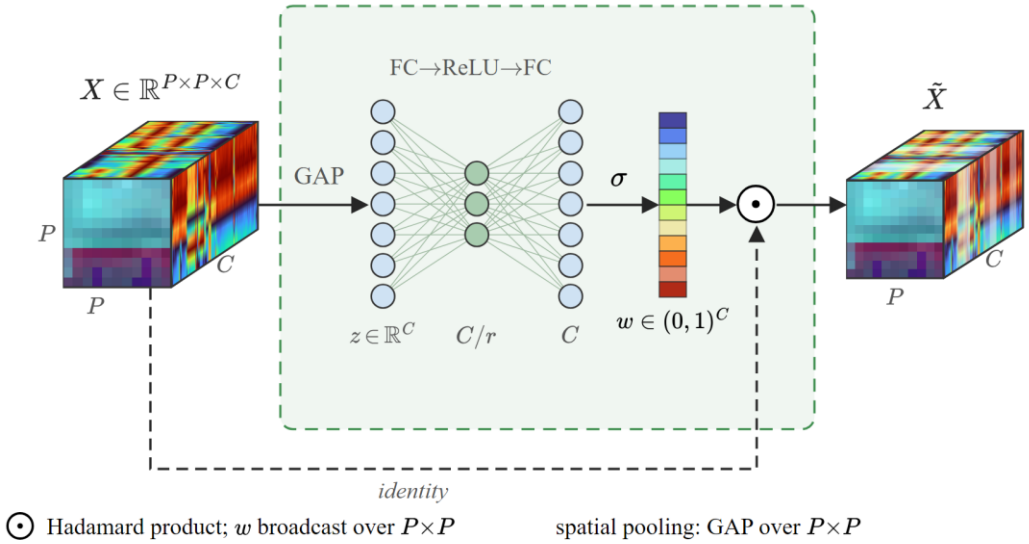

Input: X ∈ ℝ^(B×1×P×P×C) → SBAM recalibration → Dual-branch CBSA → Spectral deepening → SCFG covariance gating → Spatial refinement → ŷ ∈ ℝ^(B×C_cls)

**Stage I: SBAM band gate → Stage II: Dual 3D entry + CBSA complementary gating → Spectral ResBlocks ×2 + ECA → Stage III: SCFG covariance transition → Stage IV: Spatial ResBlocks → GAP + FC | Loss: 0.7×Spread + 0.3×Focal**

# Module 1: Spectral Band Attention Module (SBAM)

## What SBAM Does

- Operates on RAW spectral bands BEFORE any convolution
- Adaptive 3D average pooling compresses P×P spatial grid
- Squeeze MLP bottleneck (reduction r=8) learns band weights
- Sigmoid gate w ∈ (0,1)^C suppresses noisy atmospheric bands
- Hadamard product: $\tilde{X} = X \odot w$ (channel-wise reweighting)




## Key Equations & Design Insight

- $z_c = (1/P^2) \sum X(b,1,u,v,c)$ [spatial pooling]
- $w = \sigma(W_2 \cdot \delta(W_1 z + b_1) + b_2)$ [MLP bottleneck]
- $\tilde{X} = X \odot E(w)$ [band filtering]
- 
- Recalibrates BEFORE convolution (unlike SENet/CBAM)
- Only 10,225 params (3.32% of total network)

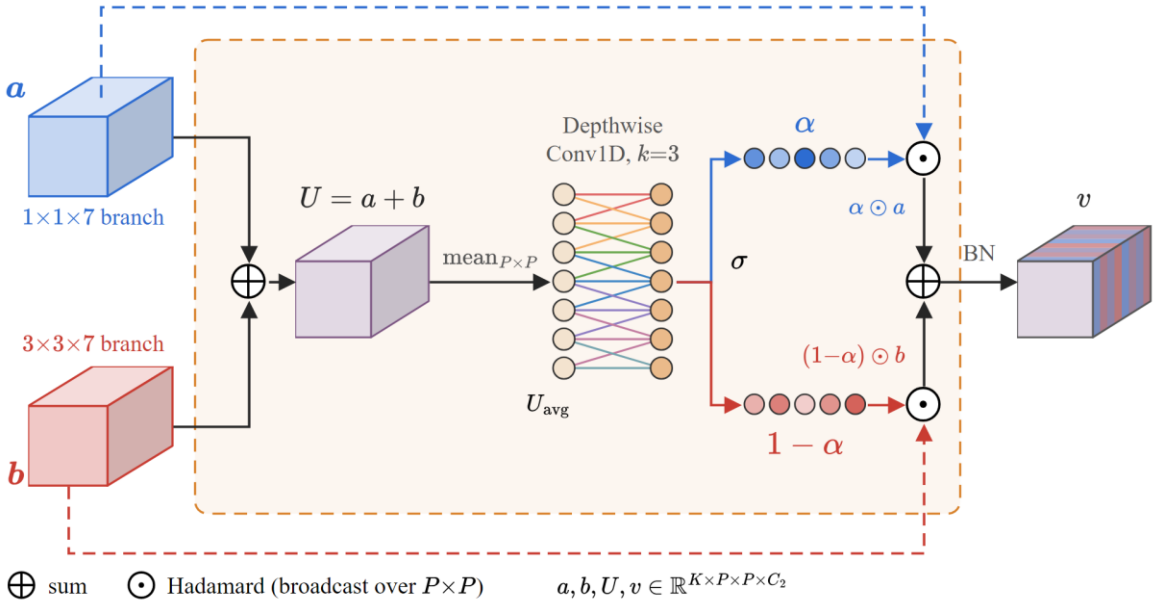
# Module 2: Complementary Branch Spectral Attention (CBSA)

## Dual-Branch Design

- Branch a: 1×1×7 Conv3D → pure spectral inter-band transitions
- Branch b: 3×3×7 Conv3D → joint spatial-spectral context
- Both: K=16 filters, spectral stride=2, BN + ReLU
- 
- Complementary Gating:
- $\alpha = \sigma(\text{Conv1D_k}(\text{MeanPool}(a + b)))$ [k=3]
- $v = \text{BN}(\alpha \odot a + (1-\alpha) \odot b)$

## Key Insight & Efficiency

- $\alpha + (1-\alpha) \equiv 1$ → ZERO redundancy between branches
- When $\alpha \to 1$: spectral fidelity dominates
- When $\alpha \to 0$: spatial context dominates
- Forces non-overlapping orthogonal representations
- Only 80 parameters (0.03%) for the gating layer
- Followed by 2 Spectral ResBlocks + ECA-3D (7,364 params)



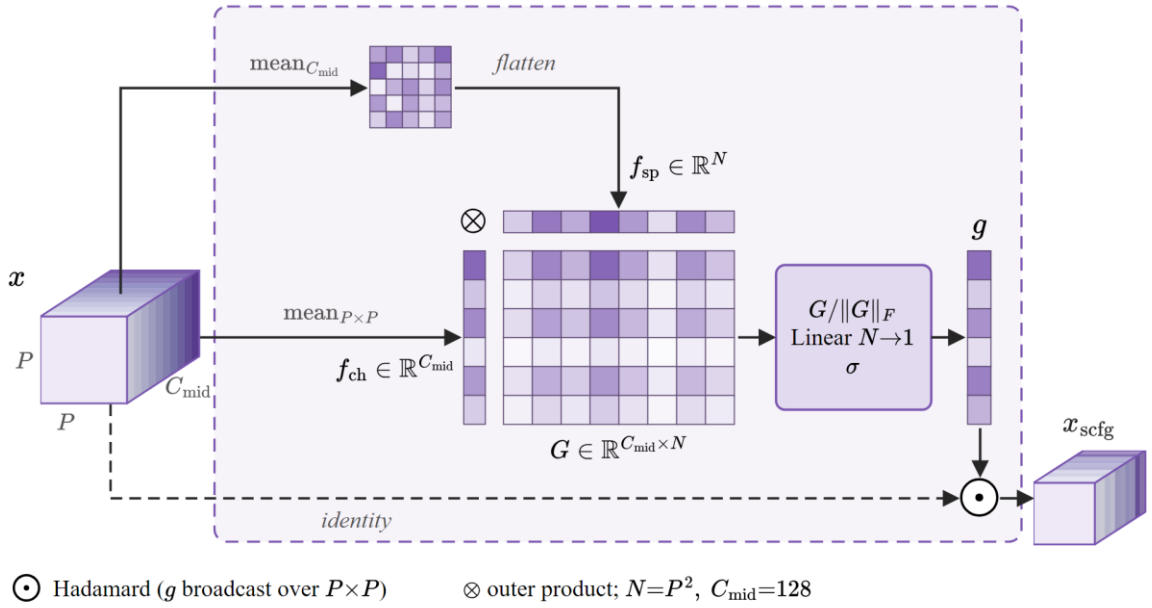
# Module 3: Second-Order Covariance Fusion Gate (SCFG)

## Covariance-Based Channel Calibration

- Transitions from spectral → spatial processing stage
- Adaptive 3D pool collapses spectral dim: $C_2 \rightarrow 1$
- 1×1×1 Conv3D expands: K=16 → C_mid=128
- 
- Second-Order Statistics:
- f_ch ∈ ℝ^(B×C): channel descriptor (mean over spatial)
- f_sp ∈ ℝ^(B×N): spatial descriptor (mean over channels)
- G = f_ch ⊗ f_sp ∈ ℝ^(B×C×N) [outer product]

## Frobenius Normalization & Gating

- $\tilde{G} = G / (\|G\|_F + \varepsilon)$ [Frobenius norm stabilization]
- $g = \sigma(W_G \cdot \tilde{G}) \in \mathbb{R}^{(B \times C_{mid})}$ [linear projection]
- $x_{scfg} = x \odot g$ [spatial modulation]
- 
- Only 82 parameters (0.03% of network)
- Captures bilinear correlations missed by first-order GAP



# Minority-Aware Hybrid Loss Function

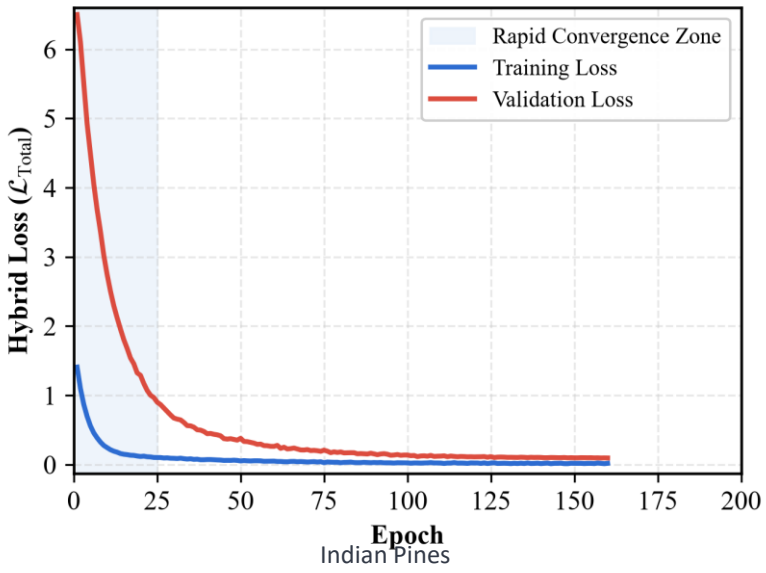
### Spread Loss (weight = 0.7)

- Dynamic margin m(r) = 0.2 + 0.7·r, scheduled over training
- Early epochs: relaxed margin (m=0.2) prevents over-penalization
- Late epochs: strict margin (m=0.9) enforces wide class boundaries

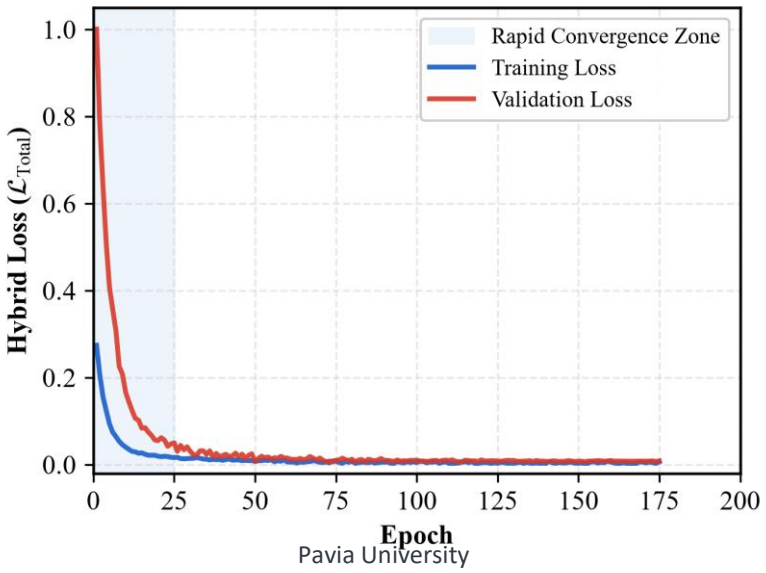
### Focal Loss (weight = 0.3)

- Inverse-frequency class weights: w_c = 1/(N_c + ε)
- Focusing parameter γ=2.0 down-weights easy-to-classify samples
- Prevents majority classes from dominating gradients over minorities

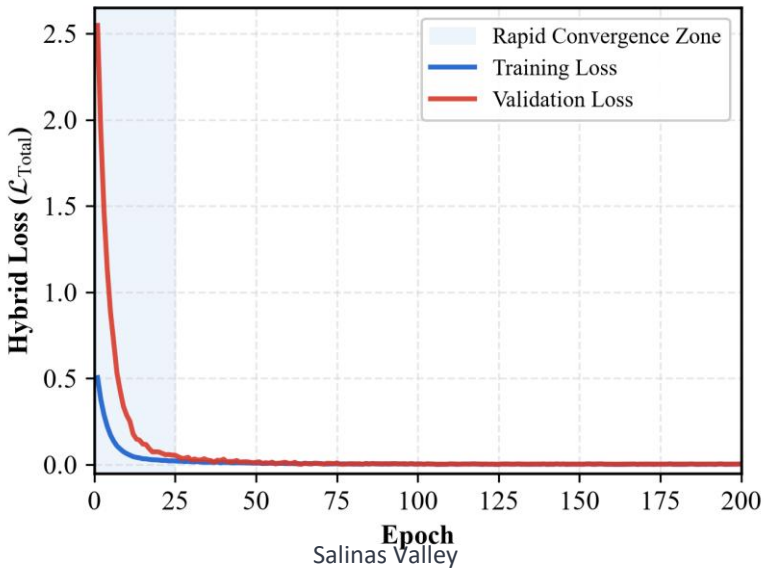
### Training & Validation Loss Convergence (200 Epochs)



Indian Pines



Pavia University



Salinas Valley

# Experimental Setup & Benchmark Datasets

| |  Indian Pines (IN) |  Pavia University |  Salinas Scene |
|---|---|---|---|
| Sensor | AVIRIS | ROSIS | AVIRIS |
| Resolution | 20 m/pixel | 1.3 m/pixel | 3.7 m/pixel |
| Size | 145 × 145 | 610 × 340 | 512 × 217 |
| Bands | 200 (filtered) | 103 channels | 204 (filtered) |
| Classes | 16 | 9 | 16 |
| Labeled Samples | 10,249 | 42,776 | 54,129 |

## Standardized Evaluation Protocol

- Split: 10% Train / 10% Validation / 80% Test
- Stratified per class (preserved proportions)
- Patch size: 11×11 spatial neighborhood
- Batch size: 32, Epochs: 200
- Optimizer: Adam, LR: $5\times10^{-4} \rightarrow 10^{-6}$
- Cosine Annealing LR schedule
- Hardware: NVIDIA Tesla T4 GPU (Kaggle)

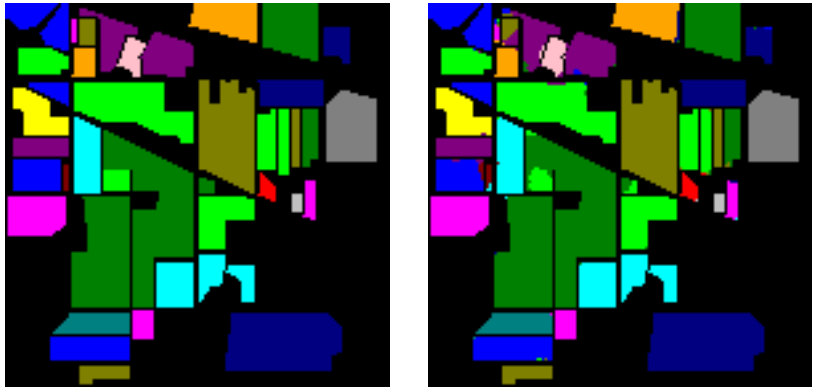
## 8 Baseline Models Benchmarked

- 3D CNNs: SSRN, A2S2K-ResNet
- Transformers: ViT, HSI-SSFTT, HyperRoPE-HiT
- Attention Networks: SSAN
- Capsule / Hybrid: DMC, CRV-MHDL
- All trained from scratch, identical protocol
- 24 total baseline runs (8 models × 3 datasets)

# Results: Indian Pines — Per-Model Comparison

| Model | OA (%) | AA (%) | κ | Time (min) |
|---|---|---|---|---|
| SSRN | **98.52** | 79.99 | 0.9831 | 6.29 |
| A2S2K-ResNet | 98.11 | 94.41 | 0.9785 | 14.89 |
| HyperRoPE-HiT | 94.07 | 89.37 | 0.9324 | 1.43 |
| HSI-SSFTT | 98.45 | 94.29 | 0.9823 | 1.05 |
| DMC | 82.55 | 64.11 | 0.7989 | 17.73 |
| ViT | 90.33 | 89.12 | 0.8895 | 1.68 |
| CRV-MHDL | 67.20 | 51.71 | 0.6222 | 1.33 |
| SSAN | 64.19 | 72.77 | 0.5979 | 14.00 |
| | | | | |
| **SGSCF-Net (Ours)** | **98.05** | **94.82** | **0.9778** | **8.80** |

## Classification Maps



Ground Truth | SGSCF-Net

## Key Finding

- SSRN tops OA (98.52%) but 0% on
- Alfalfa, Grass-mowed, Oats
- SGSCF-Net: BEST AA (94.82%)
- Alfalfa: 95.00%, Oats: 54.17%
- vs. SSRN: 0%, 0%, 0% on those

# Results: Pavia University — Urban Material Classification

| 99.82% | 99.71% | 99.76 | 12.9 min |
|---|---|---|---|
| Overall Accuracy | Average Accuracy | Kappa (κ × 100) | Training Time |



Ground Truth



SGSCF-Net

## Per-Class Highlights

- 4 classes at 100.00%: Asphalt, Painted Metal, Bare Soil, Bitumen
- Meadows: 100.00% (14,919 test pixels classified perfectly)
- Gravel: 99.76% — historically confused with Asphalt
- Shadows (947 px): 99.34% via SCFG covariance operator

## Baseline Context

- SSRN: 99.87% OA (marginally higher, 11.36 min)
- A2S2K-ResNet: 99.83% OA, 41.12 min (3.2× slower)
- SSAN: 76.82% OA — Gravel drops to 17.96%

# Results: Salinas Valley — Fine-Grained Agriculture

| 99.92% | 99.92% | 99.91 | 57.7 min |
|---|---|---|---|
| Overall Accuracy | Average Accuracy | Kappa (κ × 100) | Training Time |



Ground Truth



SGSCF-Net

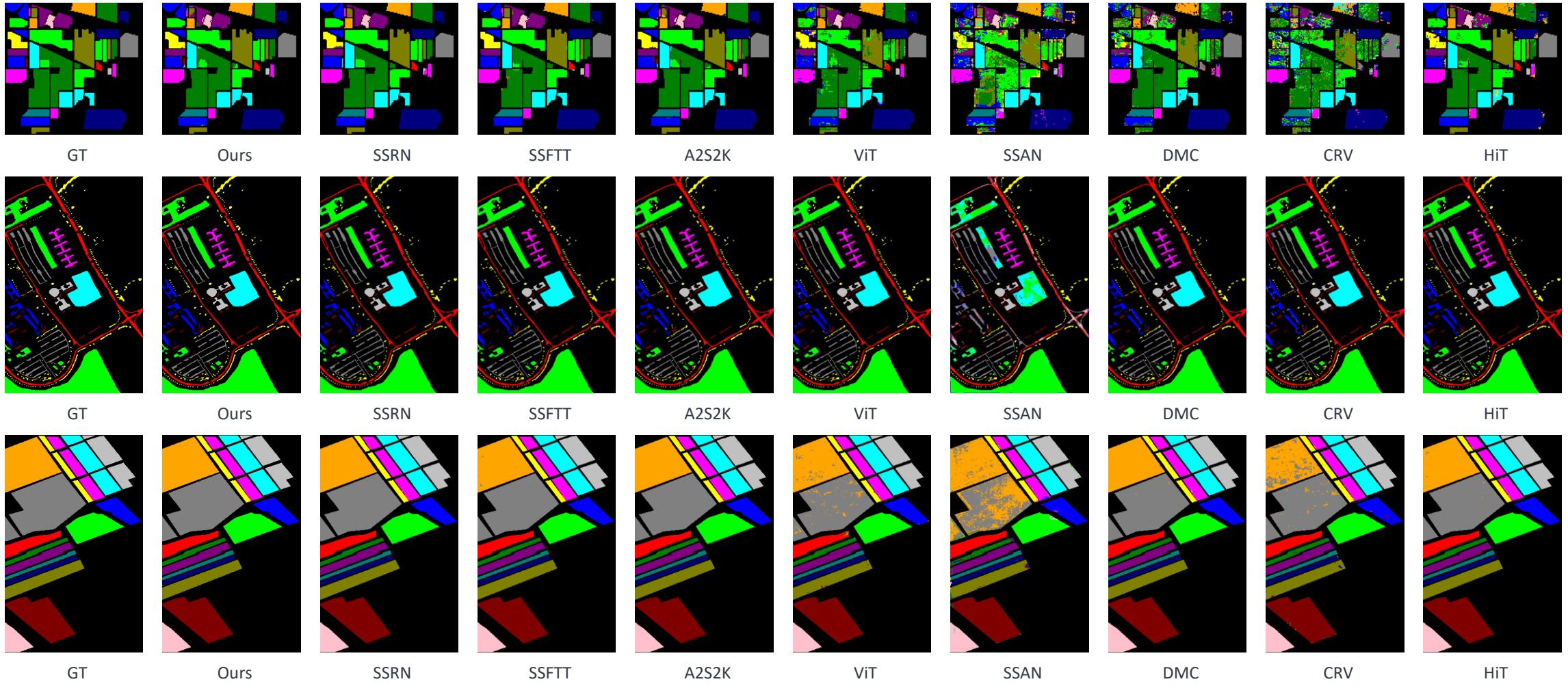
## Per-Class Highlights

- 11 of 16 classes reach 100.00% accuracy
- All 4 Romaine lettuce growth stages (4-7 wk): 99.65-100%
- Vineyard untrained: 99.73% vs. Vineyard trellis: 100.00%
- SCFG captures chlorophyll absorption band-ratio shifts

## Baseline Context

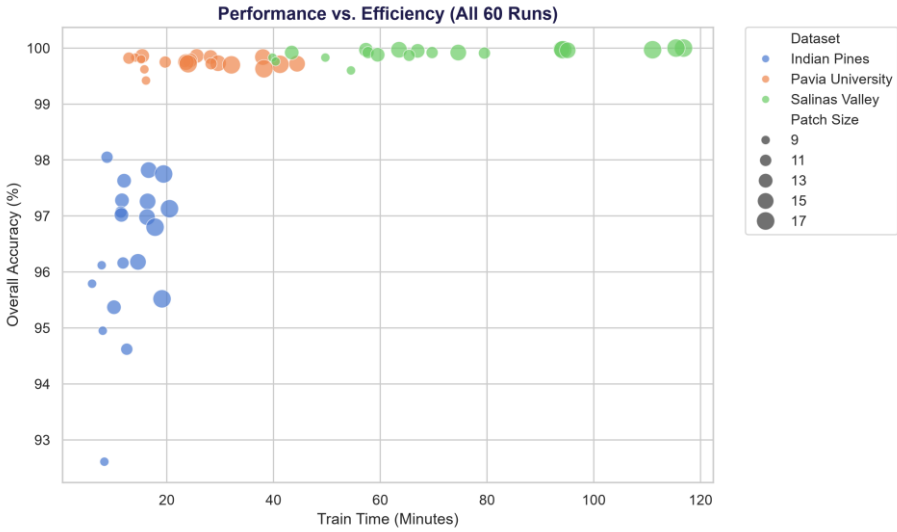
- A2S2K-ResNet: 100.00% OA (10.34 min, similar speed)
- SSAN: 88.20% OA — Grapes untrained 57.89%, collapse
- OA trend: 98.05% (IP) → 99.82% (PU) → 99.92% (SV)

# Visual Comparison: Classification Maps Across Models



# Computational Efficiency & Parameter Analysis

| Model | Params | Checkpoint | IP Time | OA (%) |
|---|---|---|---|---|
| ViT | ~610K | 2.4 MB | 1.68 min | 90.33 |
| HSI-SSFTT | ~156K | 625 KB | 1.05 min | 98.45 |
| A2S2K-ResNet | ~380K | 1.5 MB | 14.89 min | 98.11 |
| SSAN | 6.40M | — | 14.00 min | 64.19 |
| DMC | ~480K | — | 17.73 min | 82.55 |
| **SGSCF-Net (Ours)** | **308,097** | **1.2 MB** | **8.80 min** | **98.05** |

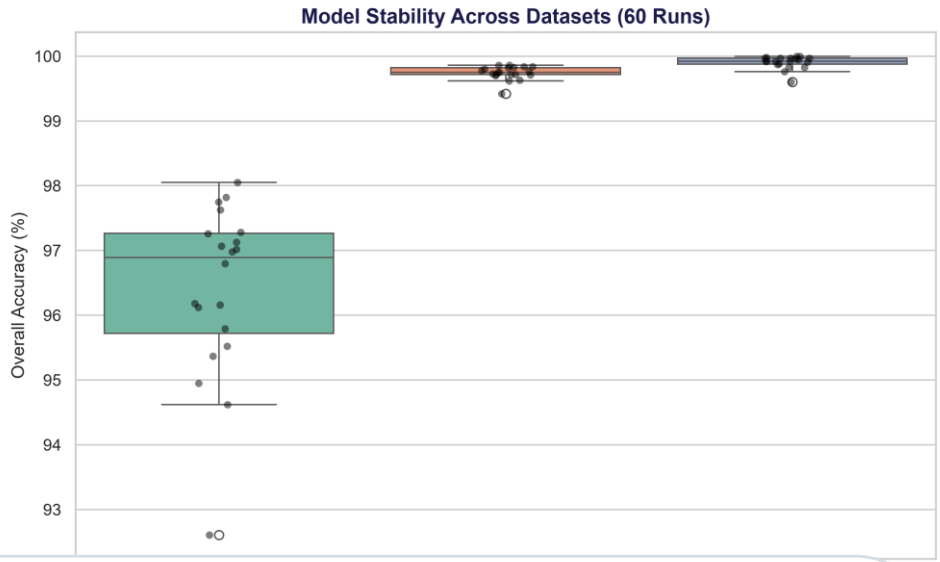



## Module-Level Parameter Breakdown (Indian Pines)

- SBAM: 10,225 (3.32%) | CBSA: 80 (0.03%) | SCFG: 82 (0.03%) → 3 novel modules = only 10,387 params (3.37% of total)
- Primary volume: Spatial 3D entry (11.98%) + Spatial ResBlocks (81.00%) — standard Conv3D backbone
- Classifier head: 272 params (0.09%) — single linear layer, no deep FC stacks

# Ablation Study & Block Depth Sensitivity

| # | SBAM | CBSA | SCFG | H.Loss | OA (%) | AA (%) |
|---|---|---|---|---|---|---|
| 1 | ✓ | × | × | × | 93.65 | 89.24 |
| 2 | ✓ | ✓ | × | × | 95.18 | 91.80 |
| 3 | ✓ | ✓ | ✓ | × | 96.44 | 93.05 |
| 4 | ✓ | ✓ | ✓ | CE | 97.26 | 93.78 |
| **5** | **✓** | **✓** | **✓** | **✓** | **98.05** | **94.82** |




## Ablation Insights

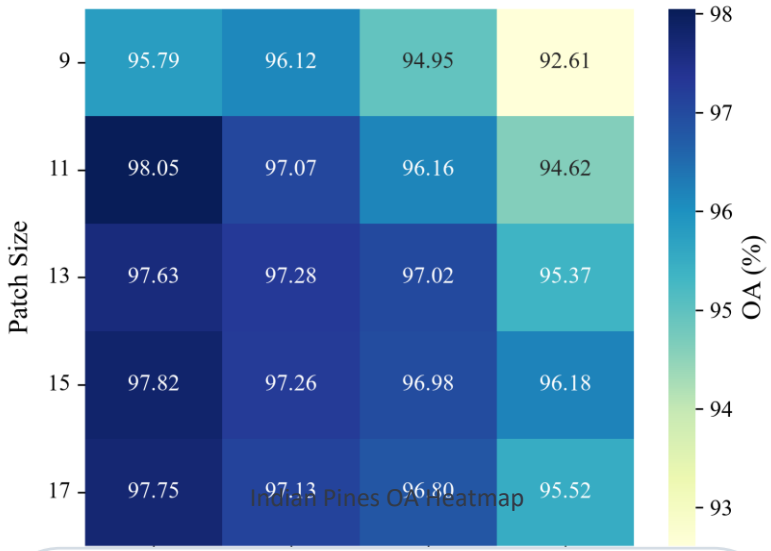
- SBAM alone lifts baseline from ~90% to 93.65% OA
- Replacing shared SE gate with CBSA: +1.53% OA
- SCFG adds +1.26% via second-order correlations
- Hybrid loss: +0.79% OA, +1.04% AA (minority protection)
- Oats class: 0% (CE-only) → 54.17% (hybrid loss)

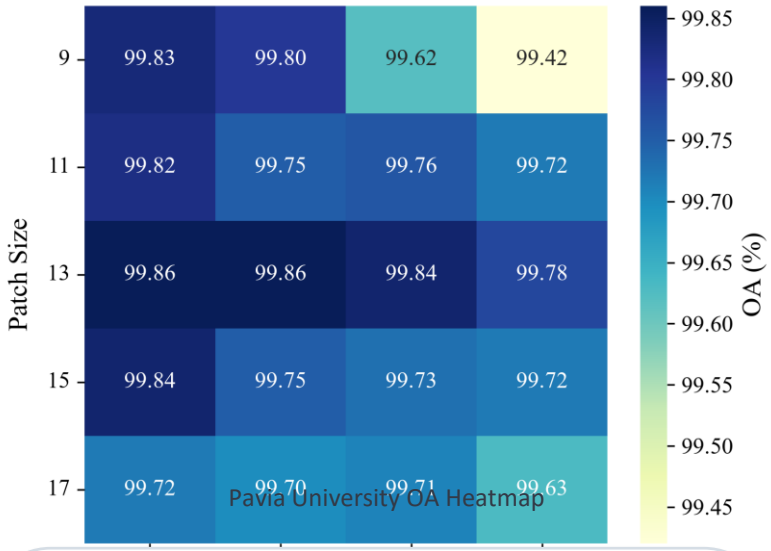
## Block Depth Sensitivity (D)

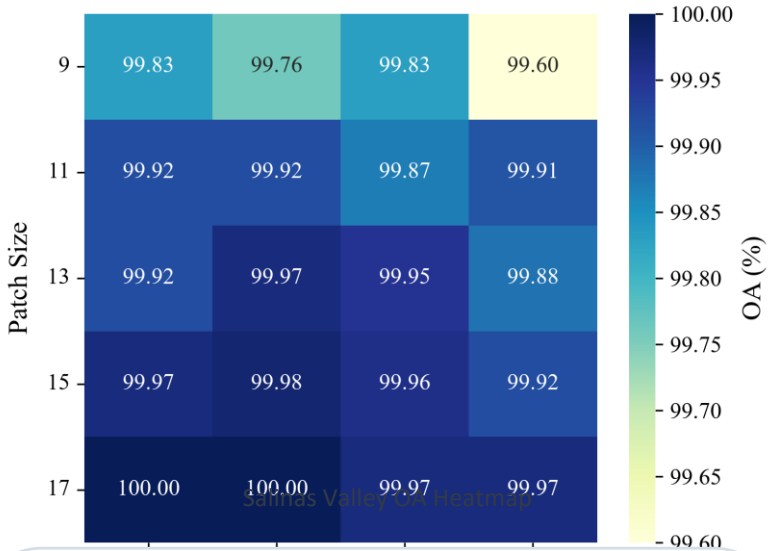
- D=1: 179K params, 95.82% OA (underfits)
- D=2: 308K params, 98.05% OA (optimal ✓)
- D=3: 437K params, 97.88% OA (diminishing returns)
- D=4: 566K params, 97.40% OA (overfitting onset)
- D=2 yields best accuracy-efficiency trade-off

# 60-Run Hyperparameter Grid Search

**5 Patch Sizes (9×9, 11×11, 13×13, 15×15, 17×17) × 4 Batch Sizes (32, 64, 128, 256) = 20 runs/dataset × 3 datasets = 60 total**




## Optimal Configuration

- Patch 11×11, Batch 32
- OA: 98.05% (IP), 99.82% (PU), 99.92% (SV)
- Consistent Rank 1 or within 0.08%

## Spatial Window Trade-off

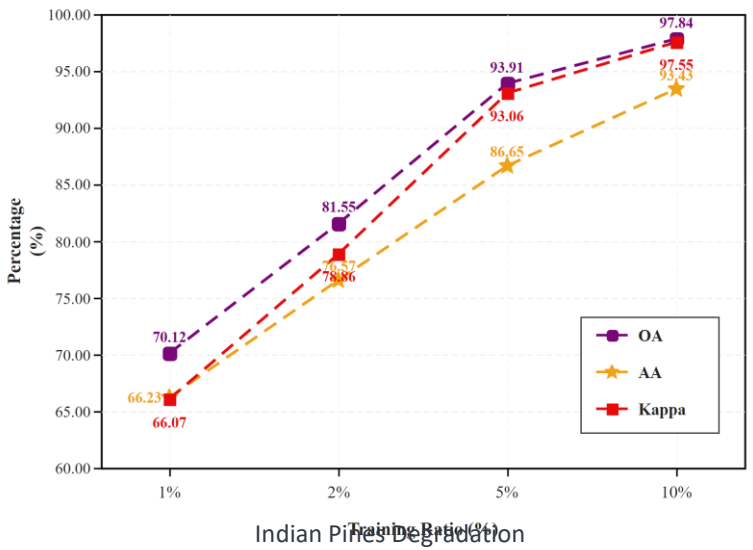
- 9×9: underfits (misses texture boundaries)
- 15×15, 17×17: overfits (cross-field contamination)
- 11×11 at 20m/pixel = 220m ground distance

## Batch Size Dynamics

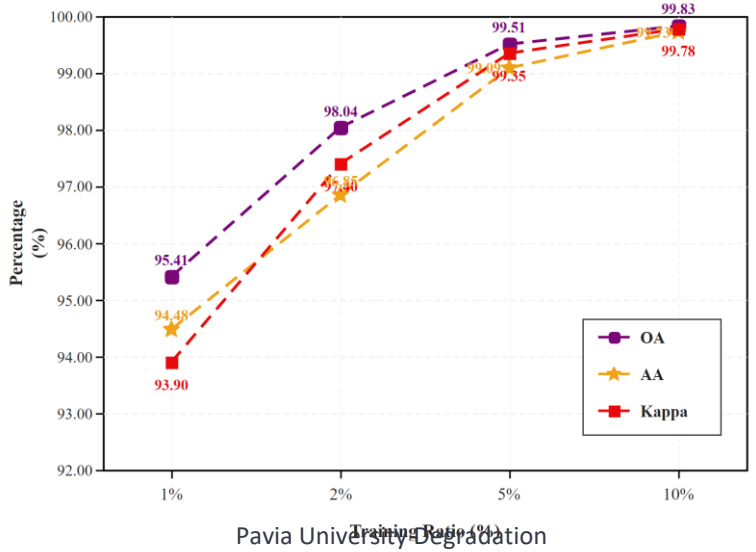
- B=32 beats B=256 by 3.43% on Indian Pines
- Small batches: higher gradient variance
- Preserves minority class decision boundaries

# Data Scarcity Stress Testing: 1% – 10% Training Ratios

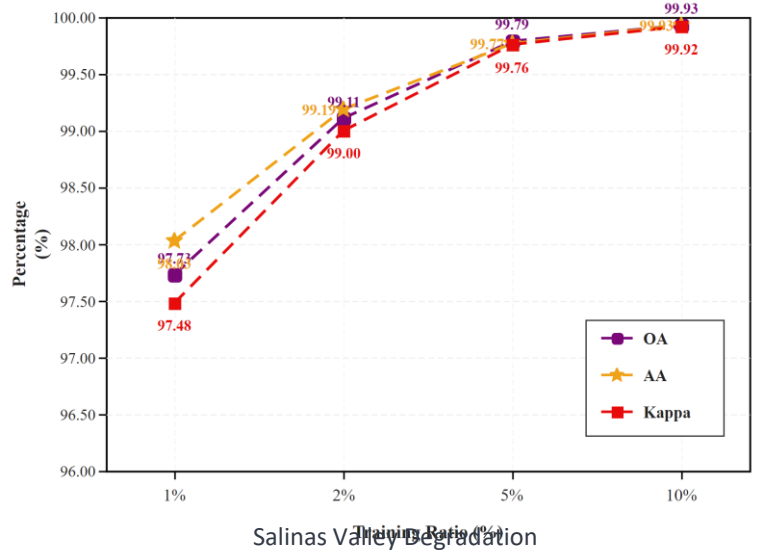
| Dataset | 1% Train | 2% Train | 5% Train | 10% Train | Drop |
|---|---|---|---|---|---|
| **Indian Pines** | 70.12% | 81.55% | 93.91% | 97.84% | -27.73% |
| **Pavia University** | 95.41% | 98.04% | 99.51% | 99.83% | -4.43% |
| **Salinas Valley** | 97.73% | 99.11% | 99.79% | 99.93% | -2.20% |



Indian Pines Degradation



Pavia University Degradation



Salinas Valley Degradation

Salinas: most resilient (−2.20%, 97.73% at 1%) | Pavia: moderate (−4.43%, 95.41% at 1%) | Indian Pines: steepest drop (−27.73%, 70.12% at 1% — Oats has 1–2 patches)

# Cross-Dataset Summary & Key Achievements

| 98.05% | 99.82% | 99.92% | 308K |
|---|---|---|---|
| Indian Pines OA | Pavia Univ. OA | Salinas Valley OA | Total Parameters |

## Project Achievements

- Designed & implemented 3 novel modules (SBAM, CBSA, SCFG)
- Best Average Accuracy on Indian Pines (94.82%) across all 9 models
- 99.82% OA on Pavia, 99.92% OA on Salinas (near-perfect)
- 120+ GPU runs: 60 grid search + 12 scarcity + 24 baseline + proposed
- O(N) linear complexity — no quadratic self-attention overhead
- 1.18s test inference on Indian Pines (vs. 135.74s SSAN)
- Complete 31-page ACM-format research manuscript drafted

## Why SGSCF-Net Matters

- Lightweight: 308K params vs. 6.4M SSAN, ~610K ViT, ~480K DMC
- Balanced: Protects minority classes where SSRN/SSAN collapse to 0%
- Generalizable: Consistent top-tier across 3 diverse HSI scenes
- Efficient: 8.80 min training vs. 17.73 (DMC), 41.12 (A2S2K)
- Modular: SBAM/CBSA/SCFG are plug-and-play for any 3D CNN
- Reproducible: Fixed 10/10/80 split, 200 epochs, public datasets
- Edge-deployable: 1.2 MB checkpoint, suitable for UAV/satellite

# Future Work & Research Directions

## 1 Hardware-Aware Quantization

INT8/FP8 tensor quantization + structured pruning for real-time inference on edge NPUs and FPGAs aboard UAVs and satellite platforms

## 2 Self-Supervised Pretraining

Masked spectral pre-training on large uncurated HSI cubes to learn transferable representations and reduce label dependency below 1%

## 3 Multi-Modal Fusion

Extend SCFG covariance mechanism to joint HSI + LiDAR classification for multi-sensor Earth observation pipelines

## 4 Journal Publication

Target IEEE Transactions on Geoscience and Remote Sensing (TGRS) or similar Q1 journal for peer-reviewed publication of this work

# Thank You

## Questions & Discussion

SGSCF-Net: Spectral-Guided Spatial Cross-Attention Fusion Network

for Hyperspectral Image Classification


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